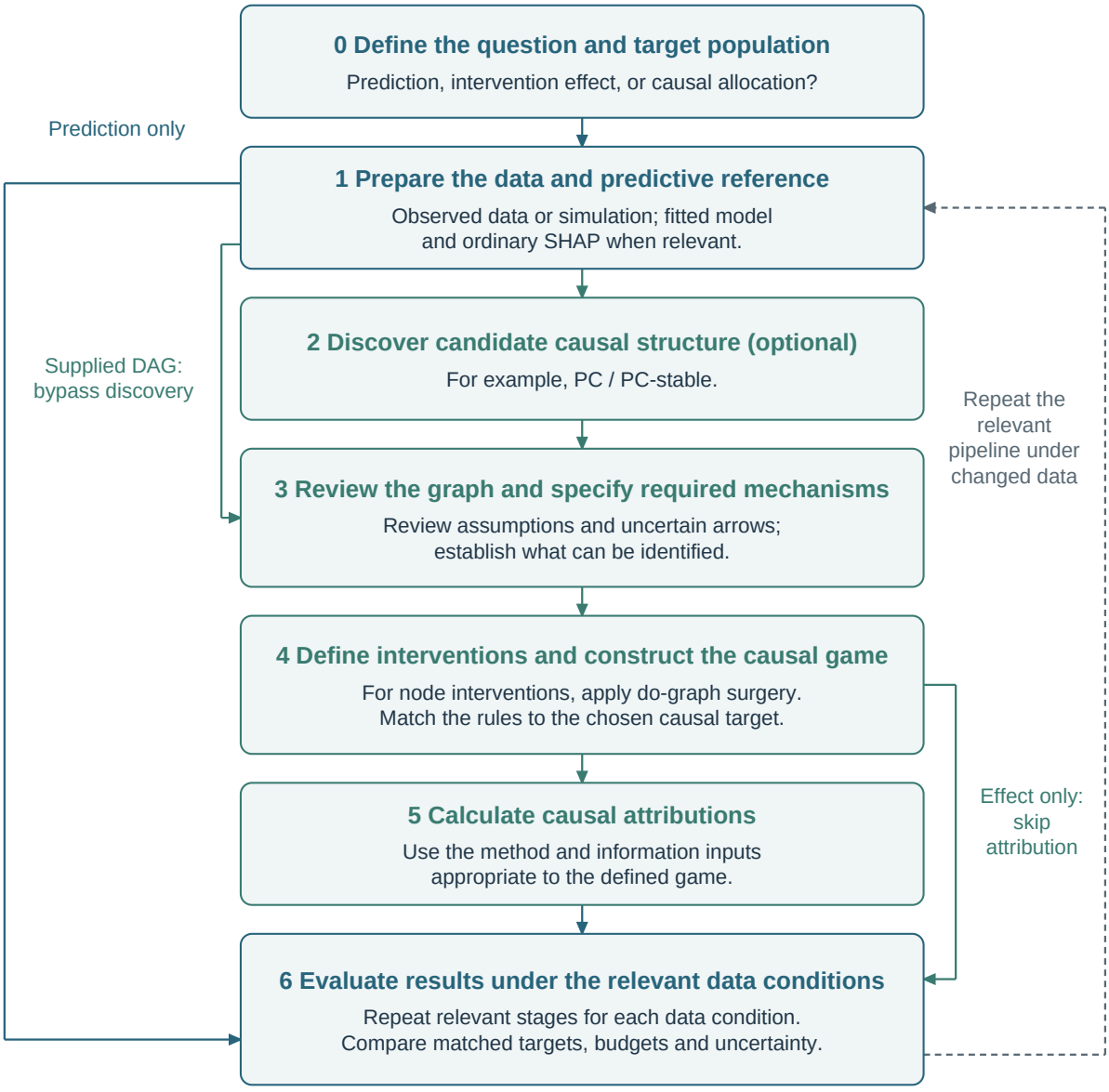
Supplementary Information

Analytical Workflow and Method Choices

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This explanatory companion follows the same Steps 0–6 as the manuscript's Methods. It distinguishes prediction, intervention effects and causal attribution, then discusses common method choices and conditional recommendations. The study's procedures and reproducibility details remain in the main manuscript. The comparisons below are selective; they do not survey every combination of method and data setting.



Supplementary Figure 1: The shared analytical sequence. Step 0 defines the question; Steps 1–6 specify the analysis. Prediction reaches evaluation directly, a supplied graph bypasses discovery, and an effect-only analysis can omit allocation. Changed data conditions require rerunning the relevant steps.

From the scientific problem to the analysis target

Define the problem and decision. Start with the health or mission question, the population, the time horizon and the decision the analysis is meant to inform. Shared definitions and explicit assumptions make interconnected risk pathways and competing outcomes discussable. Treat the DAG as a living hypothesis that can be revised as evidence changes. This perspective motivates the workflow; it is not an empirical result of the simulations.

Choose the target. A familiar prediction workflow asks how well an outcome can be forecast from available information. An intervention question asks what would change under a specified action and comparator. Causal framing can organize either question, but prediction does not require identification of an intervention effect. Move from a predictive explanation to an intervention claim only when the additional assumptions and evidence justify that move. Neither upstream position nor feature importance proves that earlier intervention is preferable; feasible actions, timing, tradeoffs and other outcomes also matter.

These framing questions guide the choices within manuscript Steps 0–1; they do not renumber the shared Steps 0–6. Encounter technical modules when a decision requires them: prediction and validation first; optional discovery if a graph is absent; graph review and identification for intervention questions; game construction and allocation only for attribution questions. The alternatives below support those decisions rather than define a required methods catalog.

One sequence for the manuscript and companion

The same step numbers identify the same operations in both documents. Each step has an explicit output; choices within it depend on the question, the data and the causal information available. The sequence supports a prediction route and a causal route without requiring every analysis to use every operation.

Step	Output and method choice
0. Define the question and target population	A prediction target, intervention contrast or allocation of a joint contrast, with population and timing specified.
1. Prepare the data and predictive reference	A declared data source or simulation, fitted reference predictor and ordinary SHAP comparator when needed.
2. Discover candidate causal structure (optional)	PC or a justified alternative produces candidate structure, retaining directions the data cannot resolve. A supplied graph bypasses this step.
3. Review the graph and specify required mechanisms	A documented graph assessment, identifying assumptions and the conditional mechanisms required by the chosen method.
4. Define interventions and construct the causal game	Declared intervention values, players, background and scale; do-surgery and identified coalition quantities where the method calls for them.
5. Calculate causal attributions	Contributions under the chosen node-, order-, weight- or edge-based allocation rule.
6. Evaluate results under the relevant data conditions	Target-specific performance, uncertainty and failure diagnostics; repeated analysis when sampling or measurement conditions change.

Prediction route. Use the reference predictor from Step 1 and assess predictive performance in Step 6. Ordinary SHAP can explain that fitted predictor; it does not establish intervention effects.

Causal route. Use Steps 2–5 as required by the target and available information. A supplied DAG enters review directly. An intervention-effect question needs identification and effect calculation, but can omit Shapley allocation. Ordinary SHAP is the study's comparator, not a prerequisite for causal inference.

The proposed detector is an optional research extension that could flag candidates for graph review; it remains unevaluated. Longitudinal analysis and cost-constrained recourse are separate extensions beyond the completed work. Their status does not add steps to this sequence.

Steps 0–1: question, data and predictive reference

Step 0. Specify the quantity before selecting a method. The manuscript's primary exposure is cumulative mission duration and its outcome is nephrolithiasis. A risk forecast, a high-versus-low mission-duration effect and an allocation of a joint intervention contrast are different targets. A hydration-policy contrast can be a separate worked intervention, with its own timing and comparator. Harrell's purposes are hypothesis testing, estimation and prediction; modeling supports all three [1]. The prediction/intervention distinction concerns the claim we need to make [2].

Step 1. Use the supplied renal topology to define a plausible system, then state the numerical mechanisms used to generate data. Distinguish the complete supplied graph (53 nodes, 83 edges), the earlier 51-node version used for full-DAG simulations, and the adapted 14-node working model. The working model adds variables and shortcuts; it is not an induced subgraph. Source relationships and numerical simulation assumptions require separate documentation. For a blinded discovery comparison, hold the generating graph and reference quantities apart from the data given to the algorithm. An oracle-assisted calculation may use them, but must be labeled accordingly.

Fit the predictive reference on those data and compute ordinary SHAP with its background distribution and output scale stated. The existing study includes random forest and XGBoost TreeSHAP, logistic-regression LinearExplainer, and model-agnostic explanations. A prespecified regression is a useful reference; additional model flexibility is conditional on sample size and response shape. A mediator receiving large predictive credit can be correct for prediction even when an upstream intervention changes the outcome.

Step 2: discover candidate causal structure (optional)

PC is the concrete starting point in this study, including the discovery component of Ng et al.'s Causal SHAP. PC removes edges using conditional-independence tests and orients those justified by its rules; it generally returns a CPDAG, representing an equivalence class rather than a unique causal DAG [3, 4]. Under the usual formulation, causal sufficiency, acyclicity, Markov and faithfulness assumptions matter. A supplied graph can bypass this stage and enter graph review directly.

Starting point or alternative	What changes, and why it might be appropriate
PC / PC-stable	PC-stable reduces order dependence in skeleton estimation. Choose tests suitable for the variable types; Gaussian tests are not a universal choice for the mixed renal data [5].
GES	A score-based comparison when its model and score assumptions are justified [6].
LiNGAM	Uses functional assumptions such as linearity and non-Gaussian disturbances to identify directions [7].
NOTEARS	A continuous-optimization alternative whose loss and model class must suit the data [8].

The mixed-data test choice belongs inside the discovery comparison. Algorithm names alone do not establish suitability. The observed disconnected-outcome PC result is retained as a diagnostic; it is not justification for silently adding an outcome edge.

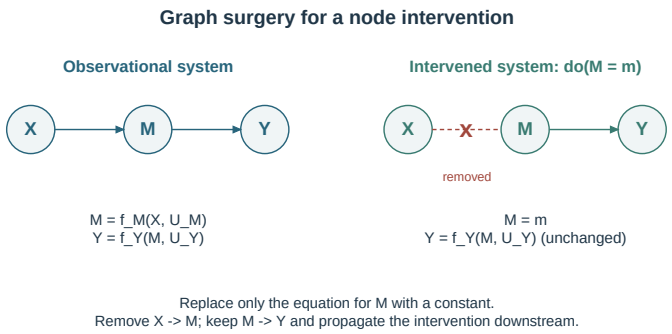
Step 3: review the graph and specify required mechanisms

Review candidate or supplied graphs against timing, biological evidence and the intended intervention. Record why each edge is retained, removed or oriented, and keep competing graphs when the evidence does not resolve them. Check whether their differences change identification of the target, not only whether their pictures look similar. Structural agreement, adjustment-set sufficiency and consequences for the target effect are distinct checks; a close-looking graph need not answer the same causal question.

Reference comparison: unrevised graph versus a documented revision under the same downstream method. **Conditional choice:** domain-informed review with an explicit ledger and retained uncertainty. For the intervention-propagation route, fit the required conditional mechanisms under the causal and identification assumptions of the chosen method, or declare that simulation mechanisms are supplied. A DAG specifies structure; it does not supply all functional forms or coefficients.

Step 4: define interventions and construct the causal game

Graph review changes our hypothesis about the observational system. *Graph surgery* instead represents a specified intervention on that system: replace the assignment equation of each intervened node by its imposed value and remove its incoming arrows. Keep the remaining mechanisms and propagate their consequences. This is the do-operator construction, not another discovery algorithm [9].



Supplementary Figure 2: For $\mathrm{do}(M = m)$ in a simple $X \to M \to Y$ chain, remove the incoming $X \to M$ arrow and replace the equation for M. The outgoing $M \to Y$ path remains. Observing $M = m$ does not perform this operation.

For a coalition C, declare whether the value is an expected outcome or a model output:

$$v_Y(C) = \mathbb{E}[Y \mid \mathrm{do}(X_C = x_C)], \qquad v_f(C) = \mathbb{E}[f(X) \mid \mathrm{do}(X_C = x_C)].$$

The two are not generally equal. The renal propagation prototype uses a fitted model output; outcome-based do-Shapley addresses a different declared game [10, 11].

In a causally sufficient, correctly specified DAG model, the post-intervention distribution can be obtained by retaining the factors for nonintervened nodes and fixing the coalition values. With observed data, identification, measurement and support must justify estimation; a drawn surgery alone does not identify an effect. Standardization or another identified estimator may replace explicit simulation when appropriate.

Reference comparison: an ordinary predictive coalition construction versus the explicitly declared interventional game. **Conditional choice:** use the latter when the target is causal allocation and its quantities are identifiable. ASV ordering and Shapley Flow have their own rules; they must not be relabeled as the same do-surgery calculation.

Step 5: calculate causal attributions

A coalition game assigns a value to each subset of variables. Once that value function and the allocation rule are fixed, compute the contributions. For a symmetric Shapley game with player set N,

$$\phi_i(v) = \sum_{C \subseteq N \setminus \{i\}} \frac{|C|!(|N| - |C| - 1)!}{|N|!} [v(C \cup \{i\}) - v(C)].$$

Changing the value function, permitted orders, players, background, output scale or aggregation changes what is being attributed. The methods below therefore occupy Step 5 without becoming interchangeable implementations.

Method	Information supplied and operation
Causal Shapley / do-Shapley	Evaluate specified interventional coalition quantities and allocate their increments. Distinguish model-output games from outcome-based games; the latter are the focus of Jung et al. [10, 11].
Ng et al. Causal SHAP	Uses PC-derived structure, IDA causal-strength information and a weighted, normalized attribution construction. It is not automatically the symmetric outcome do-Shapley game [4].
Asymmetric Shapley values (ASV)	Incorporate causal ordering through the order weights. Ordering alone does not add a mechanism for downstream intervention propagation [12].
Shapley Flow	Allocates credit to graph edges, with the explanation boundary stated. Its edge scores are a different object from node-level outcome effects [13].

Reference comparison: ordinary SHAP and the manuscript's selected causal variants, with their information inputs visible. **Conditional choice:** choose the family that matches the scientific target and the causal information available. There is no unconditional winner or verified "most cited" ranking in this guide.

For the renal study, three causal methods have run on the working subgraph. The Heskes-style working-subgraph route was blocked; the separate full-DAG propagation prototype receives known simulation mechanisms and uses graph-respecting orders. It uses supplied mechanisms and an empirical background; it does not demonstrate recovery of those mechanisms from sparse observational data.

An interventional Shapley allocation shares its specified joint contrast among players. It need not equal the effect of intervening on one player. If the question is simply the effect of one feasible intervention, report that contrast without requiring a Shapley decomposition. Action selection additionally needs feasibility, harms, costs and preferences.

Step 6: evaluate results under the relevant data conditions

The common experimental axis is to rerun selected method comparisons as data move from a clean simulated setting toward the constraints of spaceflight cohorts. Each change requires rerunning the relevant steps, including model fitting and discovery when used. The systematic degradation experiment remains proposed.

Change to the data	What to hold fixed and what to assess
Smaller samples	Keep mechanisms and targets fixed; refit the pipeline across seeds and report uncertainty and failures.
Selection	Separate entry, assignment and recording. Distinguish a changed selected-population effect from estimation or transport error.
Measurement error or missingness	Specify which nodes and mechanisms are affected; vary these separately before combining them. They are not equivalent to reducing sample size.
Mechanism or population shift	Declare what changes and the target population. Nonlinear mechanisms and out-of-distribution settings need their own checks.

Report predictive performance, graph recovery, causal-attribution accuracy and intervention-ranking agreement separately. An accuracy oracle must match the method's actual game and allocation rule. Comparing every method to a common symmetric do-Shapley target is a target-agreement study when their estimands differ. Kendall's τ against single-variable population-average intervention effects remains a separate screening-agreement diagnostic.

Reference comparison: the clean-data result with resampling. **Conditional choice:** prespecified degradation settings, repeated full-pipeline fits, matched information and computation budgets, and uncertainty tied to the claim. An evaluation-record bootstrap for one fitted model does not replace repeated training runs. Vary one consequential choice at a time where possible; the full grid of methods and settings is outside this study.

Selection in the renal example. Add eligibility E, mission assignment G and recording R alongside the biological DAG. For hydration assigned after selection, an effect among eligible assigned crew need not equal an effect in a broader source population. Weighting requires measured relevant variables, transport assumptions and support; it cannot create an absent subgroup. These indicators describe an observation process to consider when defining the target; they are not an estimated astronaut-selection model.

Current evidence. Credit shifts vary by model and pathway. The matched full-DAG ordering comparison found no detected improvement; the separate propagation prototype uses more causal information and a different budget. Working-subgraph revision rounds two and three are scripted heuristics. Human expert review, the detector, matched-game renal attribution oracles and the broader robustness comparison remain unfinished.

Scope. Keep the detector/filter as an optional, explicitly unevaluated branch. A planted mediation example can define a future detector test; it does not establish universal suppression of upstream causes. The fuller technical derivations and method comparisons remain available in the repository companion.

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